**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 097 · Lesson 097 · Little Man Computer: Branches and Loops

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does BRA do in LMC?

Q2. When does BRZ branch?

Q3. What does LDA COUNT do?

Q4. Stack Push2,Push5,Pop,Push8: final bottom-to-top?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

BRA — Always branch to label.

BRZ — Branch if ACC zero.

label — Symbolic instruction/memory address.

loop — Repeated control flow.

termination — Stop condition for loop.

## New learning

Observe — BRA always changes which instruction is fetched next; BRZ branches only when ACC is zero.

Apply — The loop's memory variable COUNT changes after SUB ONE and STA COUNT. Reload it with LDA COUNT at each iteration.

Evaluate — Trace ACC, COUNT, branch decision and outputs. Stopping before OUT when COUNT=0 prevents printing zero.

## Worked example

Start COUNT=2 instead of3. What gets printed and why?

1. Follow the reasoning — Outputs2 then1; COUNT reaches0, and BRZ branches before OUT.

# We Do · Practise with guidance

In a separate program, COUNT starts4 and SUB TWO is used, TWO DAT2. Give outputs.

Follow the reasoning

Outputs4 then2; stored COUNT becomes0, and BRZ exits before the next OUT.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Trace COUNT DAT3 loop outputs and COUNT after each SUB. [6]

### Question 2

Change COUNT DAT3 to COUNT DAT2. State outputs and explain BRZ. [6]

### Question 3

A pupil replaces BRZ END with BRA END at LOOP. Explain effect and repair. [6]

# Exit Ticket · Answer before checking

BRZ END always jumps to END regardless of ACC.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Branches unconditionally to a label.

Q2. When accumulator equals zero.

Q3. Loads COUNT's mailbox contents into ACC.

Q4. [2,8].

### You Do pages · Independent questions

Q1. Outputs3,2,1 [3]; COUNT values after decrement2,1,0 [3].

Q2. Outputs2,1 [2], COUNT becomes0 [2], BRZ exits before OUT [2].

Q3. BRA always branches [2], so no values printed [2]; BRZ tests ACC zero to allow nonzero iterations [2].

### Exit Ticket · Model answer

False. It jumps only when ACC is zero.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**